SUMANTH MALIPEDDI

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SUMMARY

Data Engineer and AI systems builder who owns production data infrastructure end-to-end at a B2B SaaS startup, covering ETL/ELT pipelines (Airflow, dbt, PostgreSQL, MongoDB), production-scale loads with zero-record-loss guarantees (811K+ records), entity resolution across heterogeneous sources, and agentic LLM pipelines, a custom MCP server driving 4,856 headless Grok agent sessions over a Postgres work queue. MS in Data Science & AI (9.5/10 GPA, IISER Tirupati). Ships complete products solo: a native iOS media player live on the App Store in 175 countries, a fintech reconciliation app, and an LLM health platform. Pivoted from civil engineering, where a self-built GCP housing-analytics pipeline over 4 years of government data catalyzed the transition. Targeting roles at the intersection of data engineering, machine learning, and AI systems.

WORK EXPERIENCE

GetMySaas (B2B SaaS Startup) Remote · Oct 2025 – Present

- *Data Science Engineer, Sole Data Infrastructure Owner* *Nov 2026 – Present*
  Associate Data Science Engineer *Oct 2025 – Nov 2026*
  - **Promoted to Data Science Engineer in 13 months**: for owning the data platform end-to-end, from ingestion (Airbyte, production MongoDB, external APIs) through dbt transformation to serving-layer marts. Scope is now expanding from individual pipelines to designing the company's production data systems.
  - **Designed and built an agentic web-research pipeline**: custom MCP server (Python FastMCP over HTTP) on a Postgres work queue with lease-based claims, driving 4,856 headless Grok agent sessions (2.2B tokens) across 219,021 vendors; least-privilege agent DB role, server-side validation of every answer, locked-down agent tool policy, and per-session token/cost accounting.
  - **Raised data-platform profile coverage to 95,089 vendors (43.4%)**: 259,889 profile links across 5 platforms by combining LLM agents with 6 search APIs benchmarked head-to-head; every agent link verified by fetching the page after audits found 20–44% of early agent links wrong.
  - **Running a 173,533-page fetch on EC2 into an S3 data lake**: per-site fetch strategy chosen by measured tests (Playwright, FlareSolverr, residential + gateway proxies); verified offload every 3h with Standard-IA/Glacier lifecycle and hourly Zoho Cliq alerting; diagnosed an IPv6/DNS64 fault that made S3 transfers 557× faster.
  - **Led recovery and postmortem of a production MongoDB outage**: disk exhaustion took `mongod` down for 11.7h; restored via oplog replay, verified all 52 collections, then shipped disk/RAM guards, preflight checks and health alerts so it cannot recur.
  - **Own the company's canonical market taxonomy**: 166 categories, architect-approved v3.6.0; classified 423,163 Product Hunt products against it with zero record loss; designed the governance layer: schema validators, machine-parseable routing overrides, and watch registers with falsifiable review triggers.
  - **Engineered production-scale loads with zero-record-loss guarantees**: loaded 811,548 person records from production AWS MongoDB over a load balancer that kills long-lived cursors, using chunked `_id` pagination with retry; staging tables + count-reconciliation gates + atomic content swap (content-swap over rename because Postgres views bind by OID).
  - **Built golden-record and outreach marts**: `dim_companies` (48,729) and `dim_vendor_contacts_outreach` (792,816) with person-level dedup and honest-schema design (no synthetic identity keys, `is_deliverable` gated FALSE until independently verified); 10/10 dbt tests passing.
  - **Implemented the dbt transformation framework**: 12+ models across staging/intermediate/marts, 32 schema and custom data tests, SCD Type 2 snapshots, seed-driven configuration (domain overrides, country normalization), and 5 reusable macros; deployed in a 6-service Docker Compose stack; Airflow orchestration via Cosmos in progress.
  - **Designed multi-source entity resolution**: multivariate match-candidate scoring across 6 heterogeneous sources (5-dimension confidence model), 14-column source-attribution provenance, dynamic regex revenue parsing, and automated same-name-different-domain quality detection.
  - **Built multi-agent LLM research systems as production tooling**: workflows up to 54 parallel agents (647 web lookups) with adversarial researcher-vs-skeptic verification, structured outputs, resume-from-cache, and model-tiering for cost control; a quantitative inclusion framework (3 hard gates, 10 weighted dimensions) turned research into architect-approved decisions.

- **Engineering Secretary, Project & Data Analytics** Tirupati, India
  Project Engineer *Oct 2019 – Jun 2024*
  - **Led quantitative project analytics**: managed construction data across 200+ residential units, including material consumption tracking, cost variance analysis, schedule adherence metrics, and workforce productivity models; reduced cost overruns by 15%.
  - **Cross-functional stakeholder management**: coordinated architects, contractors, government bodies, and clients; managed teams of 20–50 across concurrent projects, translating technical specifications into business decisions.

KEY PROJECTS

- **Housing Data Analytics: GCP Data Pipeline (GitHub)** Mage AI, BigQuery, GCS, Power BI
  Solo: career-pivot project on 4 years of government housing data *Apr 2024*
  - **Mage AI orchestration**: built the ETL pipeline as Mage AI loader, transformer and exporter blocks with inline block tests, self-hosted on a GCP Compute Engine VM, extracting from Cloud Storage and exporting to BigQuery via Mage's `io_config`.

- **Dimensional modeling**: curated records gathered from revenue, housing and municipal administration departments into a star schema (fact table + 6 dimensions: allotment, contract, in-charge, PMAY, applicant, personal) feeding Power BI stakeholder dashboards.

- **MaxMediaPlayer: Native iOS Video Player (Live on the App Store)** Swift 6, VideoToolbox, StoreKit 2
  Solo: custom playback engine, App Store release across 175 countries *Jul 2026*
  - **Systems-level media engineering**: wrote a pure-Swift Matroska/WebM (EBML) demuxer from scratch feeding a bounded-memory streaming engine, using memory-mapped reads into Apple's hardware decoders (AV1/HEVC/H.264 via VideoToolbox) playing 60GB files with instant start; O(1) seeking via Cues index.
  - **Release and ops automation**: automated the entire App Store operation via the App Store Connect API: subscription/IAP creation across 175 territories, localized listings in 5 languages, and review submission as a CI-style pipeline; passed App Review including LGPL-2.1 compliance.
  - **Engineering rigor**: 327 unit tests on the pure-logic core running on macOS + Linux CI under Swift 6 strict concurrency, in a Tuist-managed monorepo (∼20 SwiftPM packages); fixed field-reported beta issues across 9 builds in 3 weeks.

- **PaySure: Payment Reconciliation for Gig Workers** Next.js 16, TypeScript, SQLite/Drizzle
  Solo: full-stack fintech app from self-written product spec to ship *2026*
  - **Full-stack build**: 12 routes, 59 unit tests, OAuth 2.0, PDF/CSV export; converts Google Calendar events into employer-format payout statements and reconciles them against actual payments, flagging underpaid hours and TDS deductions.
  - **Fintech-grade correctness**: timezone-correct pay-period engine for India's 26th–25th cycles (UTC/IST boundary tests), integer-paise money arithmetic, three-bucket reconciliation, and TDS-aware (§194J) verification with Form 26AS ledger.
  - **Security-first integration**: OAuth read-only scope with AES-256-GCM-encrypted refresh tokens and CSRF state validation; calendar descriptions deliberately never stored; from-scratch ICS parser with RRULE expansion as no-OAuth fallback.

- **AharaOS: Condition-Aware Health OS** FastAPI, Supabase, Gemini + Claude, SwiftUI
  Solo: LLM product with multi-provider failover, deployed on Fly.io *2026*
  - **AI health platform**: 120+ REST endpoints, 40 SQL migrations, RLS on all 19 tables; generates condition-aware nutrition protocols with medical-condition filters as hard gates before macro ranking; multimodal LLM extraction of 92 markers from lab-report PDFs.
  - **Multi-provider LLM engineering**: Pydantic-schema structured outputs with automatic Gemini→Claude failover (validated during a real provider outage), per-user cost caps with a DB-driven pricing ledger, and defense-in-depth security (SSRF guard, CSP+nonce, rate limiting) behind 6 CI gates.

TECHNICAL SKILLS

- **Data Engineering**: Apache Airflow (Cosmos/Astronomer, custom operators), **Mage AI** (pipeline blocks, BigQuery export), dbt (tests, snapshots, seeds, macros), Airbyte, web scraping at scale (Playwright, curl_cffi, BeautifulSoup), ETL/ELT design, entity resolution, medallion architecture, SCD Type 2, zero-record-loss load patterns, Kafka (fundamentals; hands-on in progress)
- **Programming & SQL**: Python (Pandas, psycopg2, FastAPI, Pydantic), SQL (PostgreSQL, BigQuery: window functions, CTEs, JSONB, `DISTINCT ON`, query optimization), MongoDB (pymongo), Swift 6, TypeScript
- **AI / ML**: Agentic pipelines (custom MCP servers with FastMCP, headless Grok agents, tool-permission lockdown), multi-agent LLM orchestration (adversarial verification, structured outputs, model tiering, cost engineering), Gemini, Claude & Grok APIs, Claude Code, TF-IDF similarity, quantitative evaluation frameworks; ML/DL foundations (Scikit-learn, TensorFlow, statistical modeling)
- **Cloud & Infrastructure**: AWS (S3, EC2, IAM), GCP (Compute Engine, Cloud Composer, BigQuery, GCS), Snowflake, Docker & Docker Compose, GitHub Actions CI/CD, Fly.io, Linux, Git/GitHub PR workflow
- **Data Quality & Governance**: dbt tests, honest-schema design, count-reconciliation gates, validation frameworks, taxonomy governance, pipeline monitoring and alerting, incident postmortems

EDUCATION

- **Indian Institute of Science Education and Research (IISER)** Tirupati, India
  Master of Science – Data Science & Artificial Intelligence; GPA: 9.5/10.0 *Jul 2024 – Jul 2025*
  - **Coursework**: Machine Learning, Deep Learning, NLP, Statistical Modeling, Big Data Analytics, Database Systems, Cloud Computing, Data Structures & Algorithms

- **SASTRA University** Thanjavur, India
  Bachelor of Technology; GPA: 8.54/10.0 *Jul 2014 – May 2018*

CERTIFICATIONS

- **dbt Fundamentals**: dbt Labs: data modeling, testing, documentation, materializations
- **Apache Airflow 3 Fundamentals + DAG Authoring**: Astronomer: orchestration, TaskFlow API, advanced patterns
- **Snowflake Hands-On Essentials**: Snowflake: data warehousing, SQL analytics
- **AWS Cloud Practitioner Essentials**: AWS: cloud architecture, core services

LEADERSHIP & COMMUNITY

- **Teaching Assistant, Data Structures, Python & AI/ML** Masai School
  Technical Mentor, teach DSA, Python, and ML to aspiring engineers *2025 – Present*

- **Data Science Learning Community, @the.intlovert** 4300+ learners
  Built educational platform, 50+ mentoring sessions via Topmate; 1,100+ day daily learning streak *Aug 2025 – Present*